

MICROBIOLOGY: Bacteria I and II - Structure and Physiology

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Session Objectives

1. Compare and contrast Gram positive and Gram-negative bacterial cell envelop
2. Identify and state the importance and significance of the Gram stain as a bacteriological tool
3. Identify exceptions to the nomenclature of the Gram reaction (i.e. Wall-less bacteria, Acid-fast bacteria)
4. Determine the role of virulence factors such as capsules and LPS (endotoxin) in the pathogenesis of bacterial infections
5. Identify bacterial endospores and distinguish their unique properties
6. Identify and distinguish between medically significance spore-forming bacteria
7. Distinguish between different modes of horizontal gene transfer in bacteria and the significance in a clinical setting
8. Distinguish the different growth environments of bacteria and the significance in a clinical setting

Lecture Outline

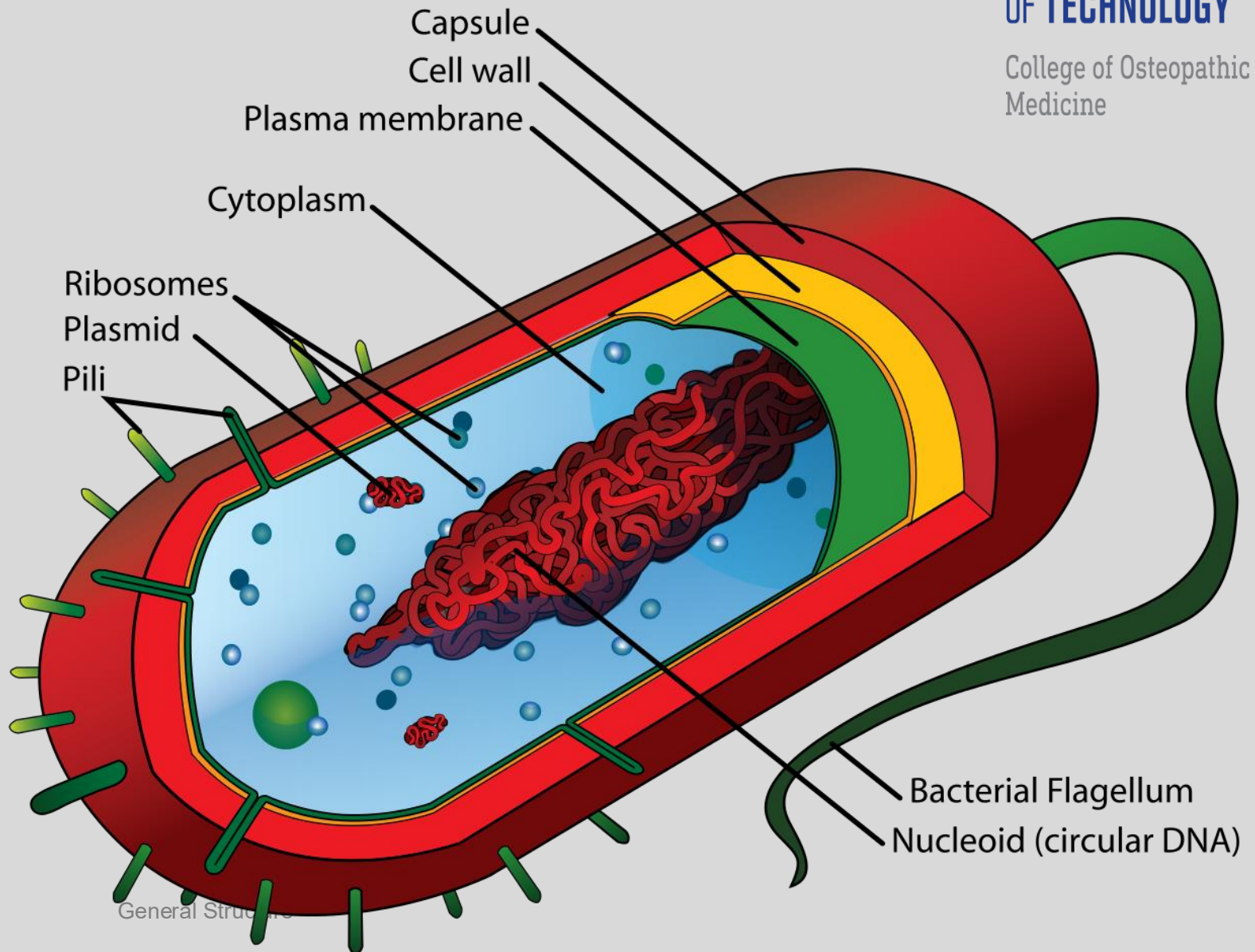
- Basic structural terminology
- Basic bacterial cell structure
- The Gram stain
- The bacterial cell wall
- Capsules
- The Acid Fast stain

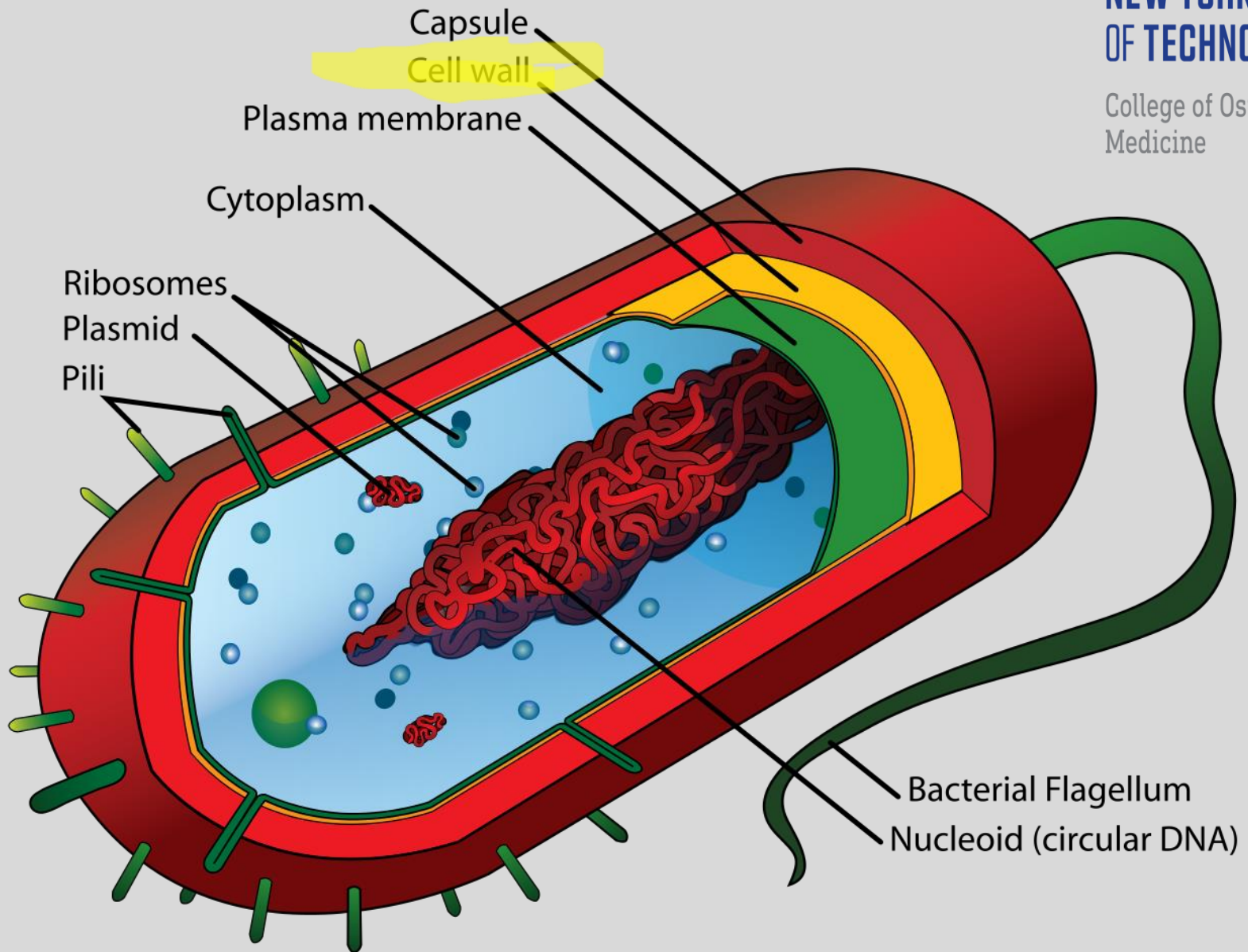
Basic terms

- Cell envelop:
 - a. Cell membrane,
 - b. Cell wall

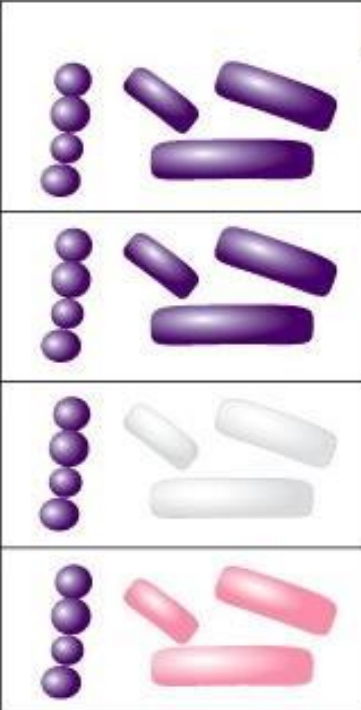
and if present:

 - c. Periplasm,
 - d. Outer membrane,
 - e. Capsule,
 - f. Pili, and
 - g. Flagella

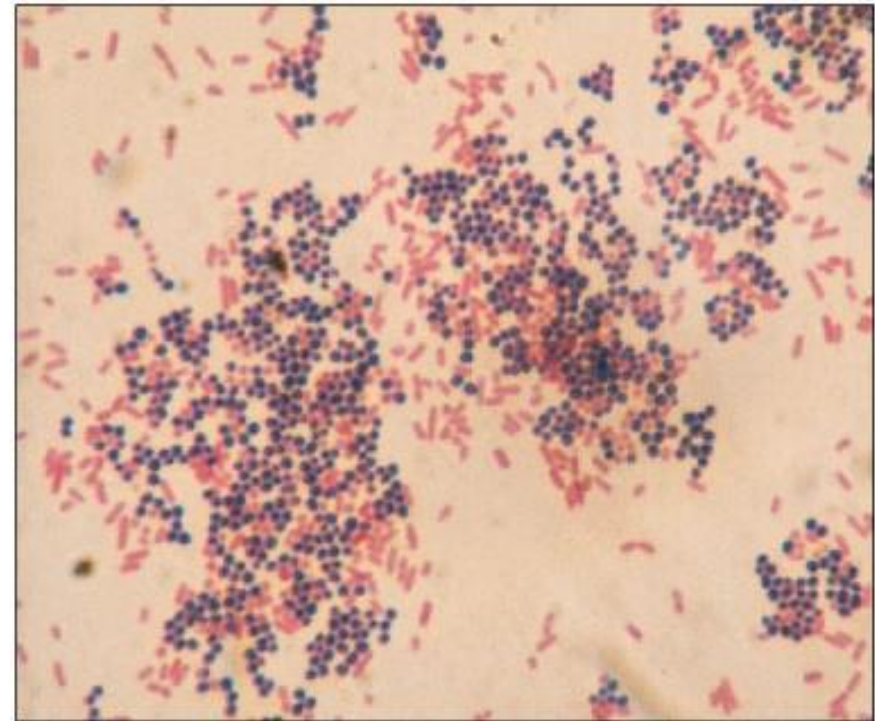




The Gram Stain

	Steps in Staining	State of Bacteria
	Step 1: Crystal violet (primary stain)	Cells stain purple.
	Step 2: Iodine (mordant)	Cells remain purple.
	Step 3: Alcohol (decolorizer)	Gram-positive cells remain purple; Gram-negative cells become colorless.
	Step 4: Safranin (counterstain)	Gram-positive cells remain purple; Gram-negative cells appear red.

(a)



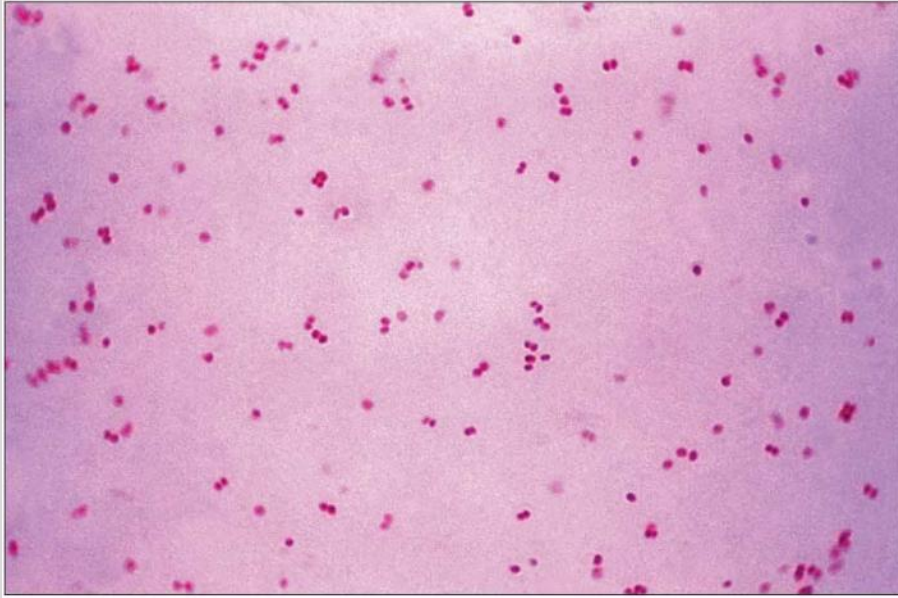
(b)

10 μ m

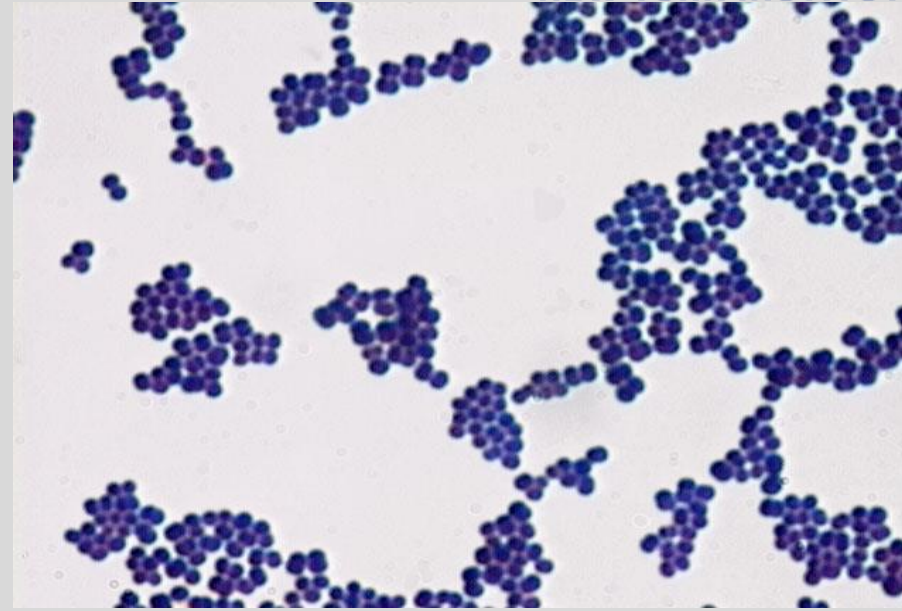
Source: The McGraw Hill companies

Gram Stain ID Power

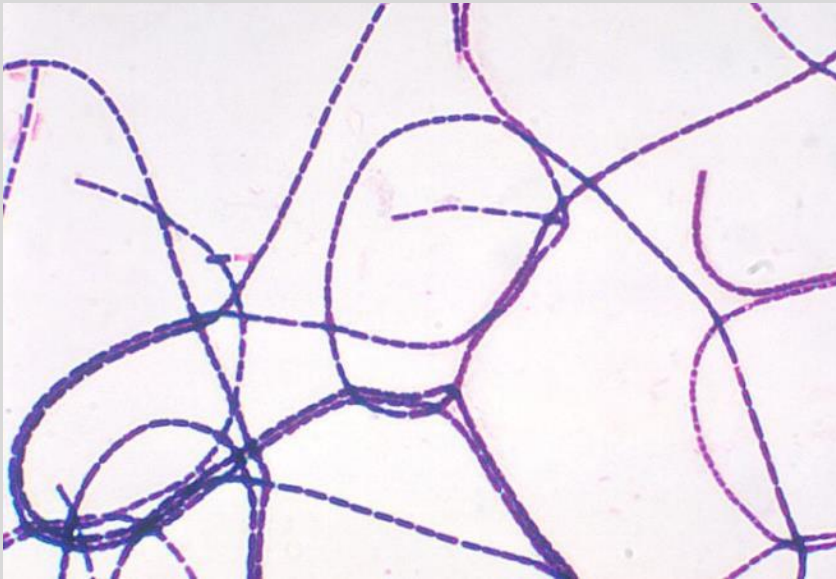
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<http://www.psychiatrictimes.com/sites/default/files/peds/1383925.png>



[https://homepages.wmich.edu/~rossbach/bios312/LabProcedures/Gram pos](https://homepages.wmich.edu/~rossbach/bios312/LabProcedures/Gram%20pos%20ID%20Power.pdf)

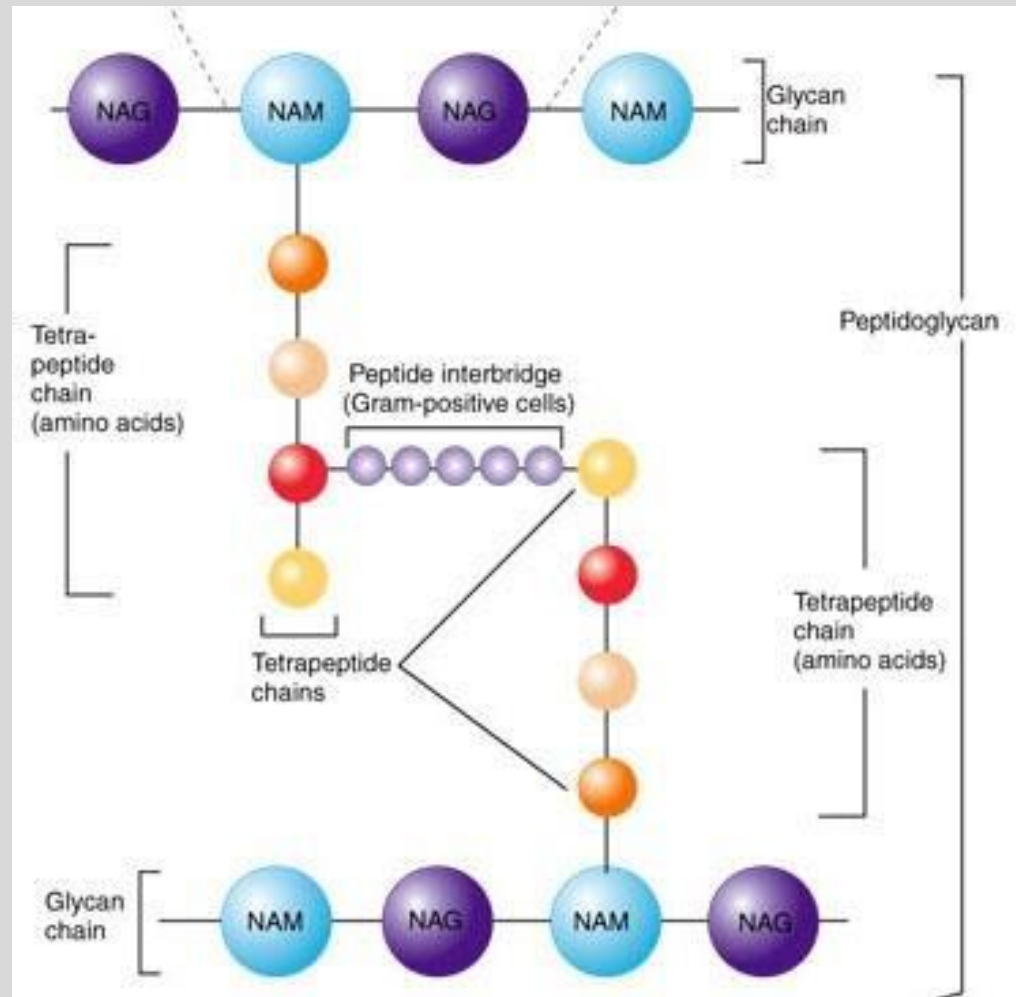


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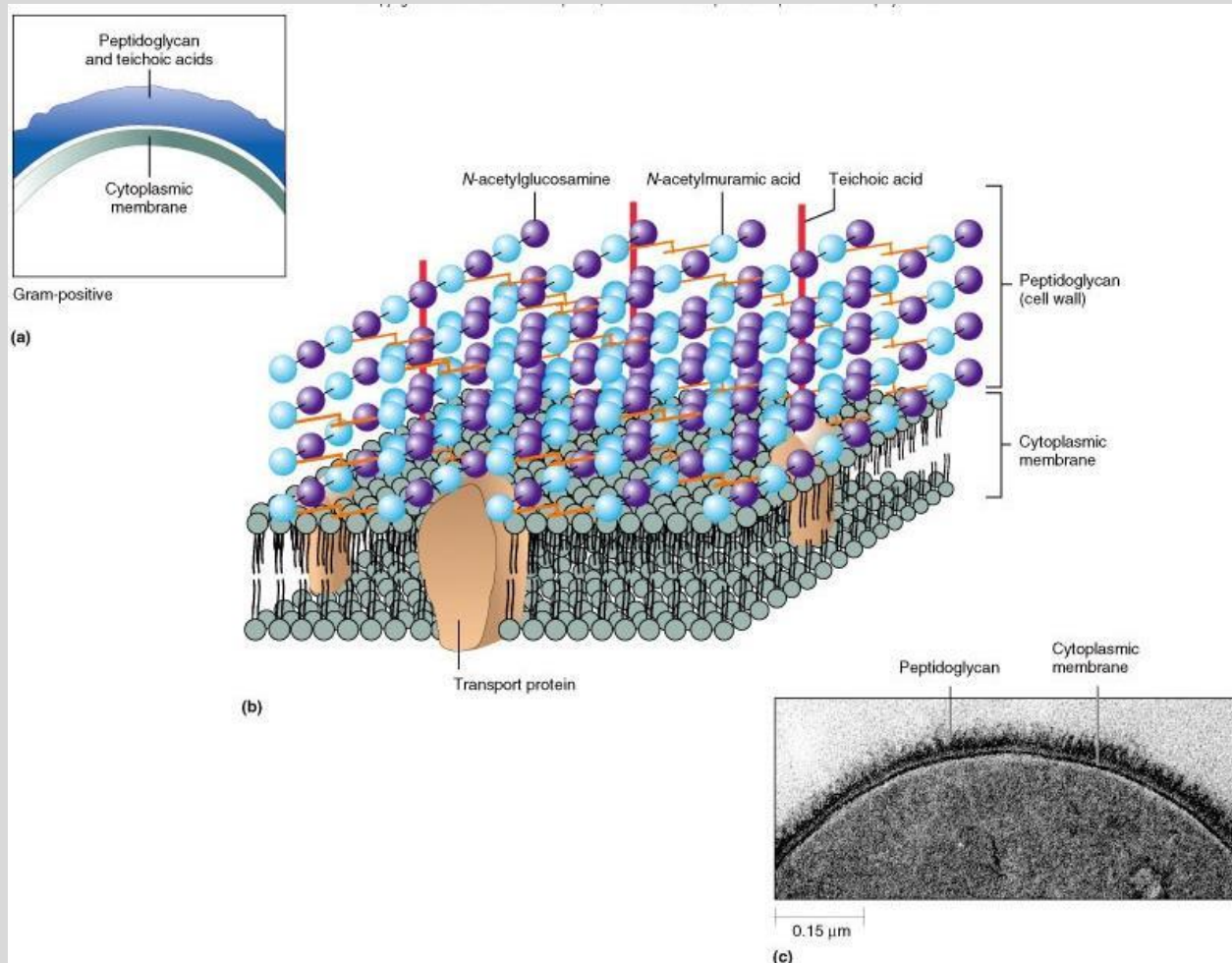
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The Cell Wall



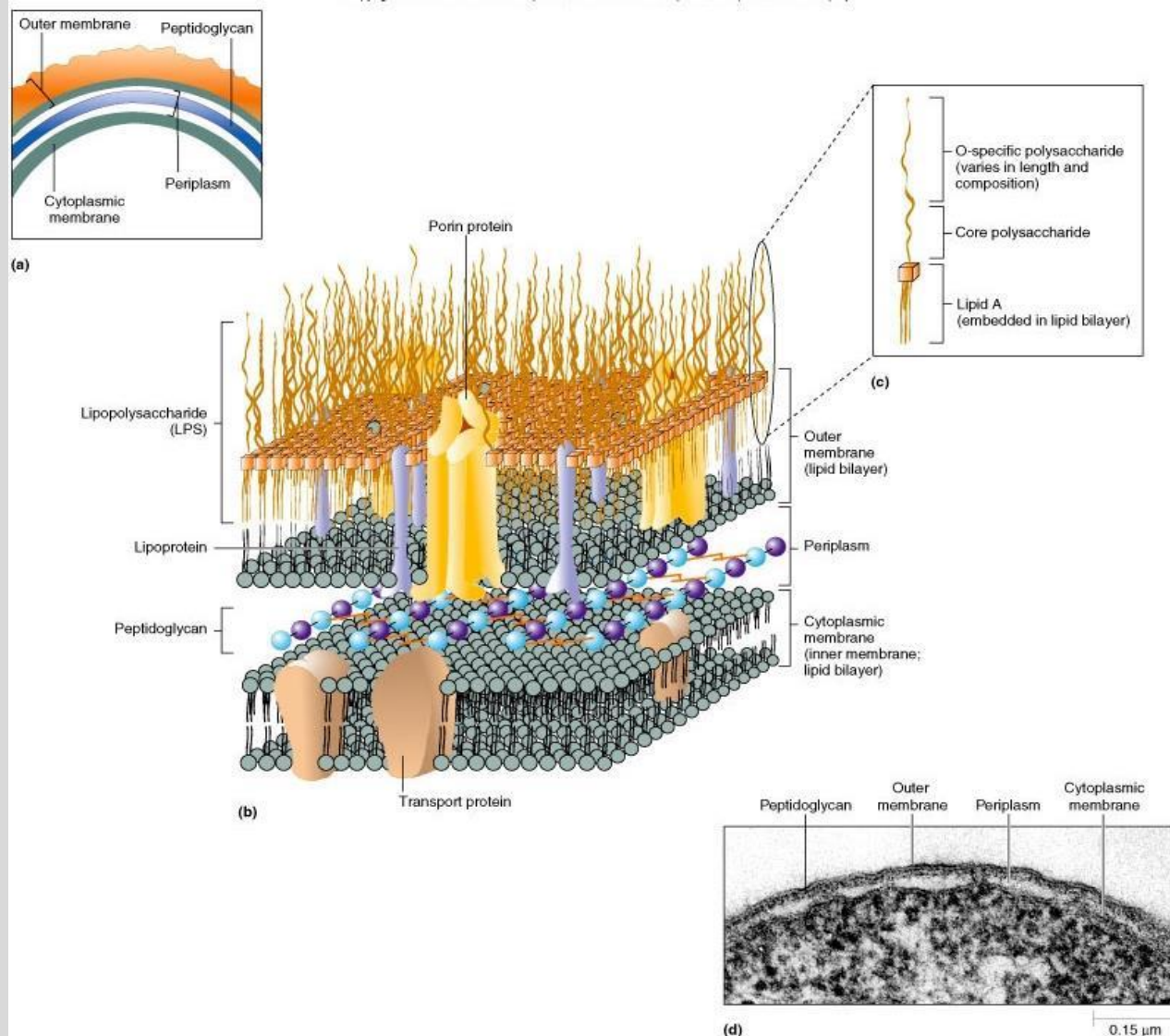
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The Gram +ve cell wall



Source: The McGraw Hill companies

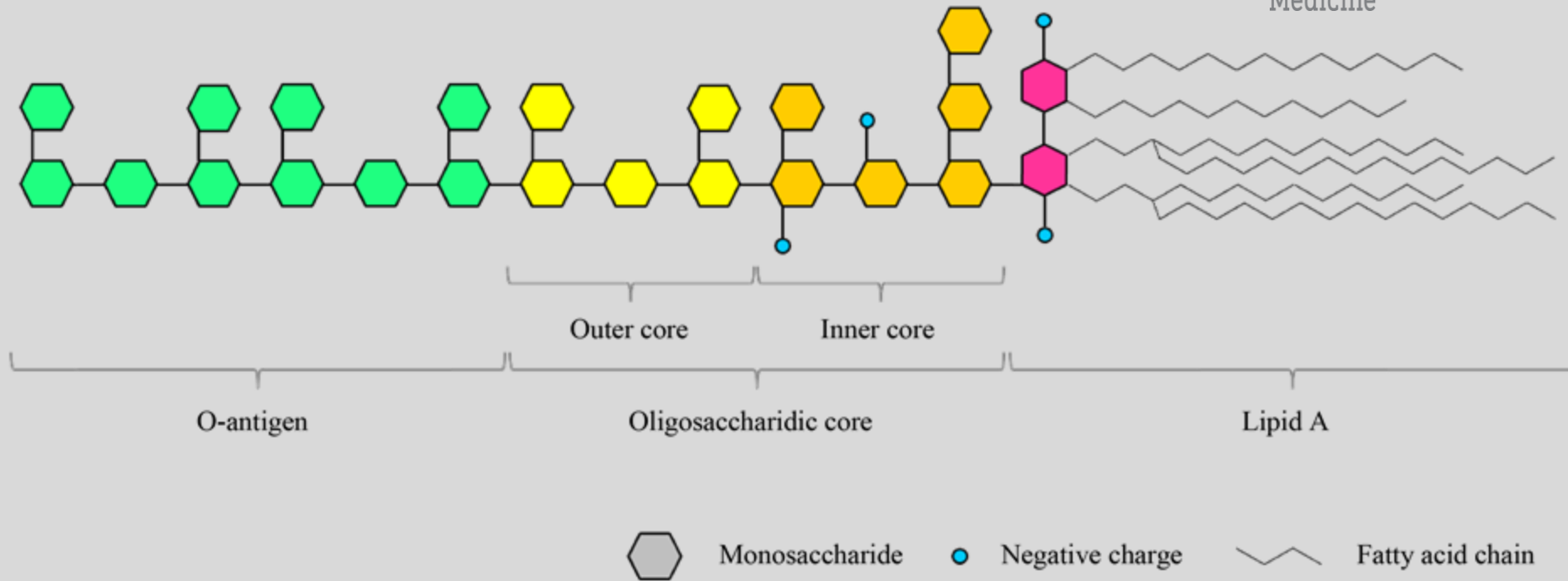
The Gram -ve cell wall



Lipopolysaccharide

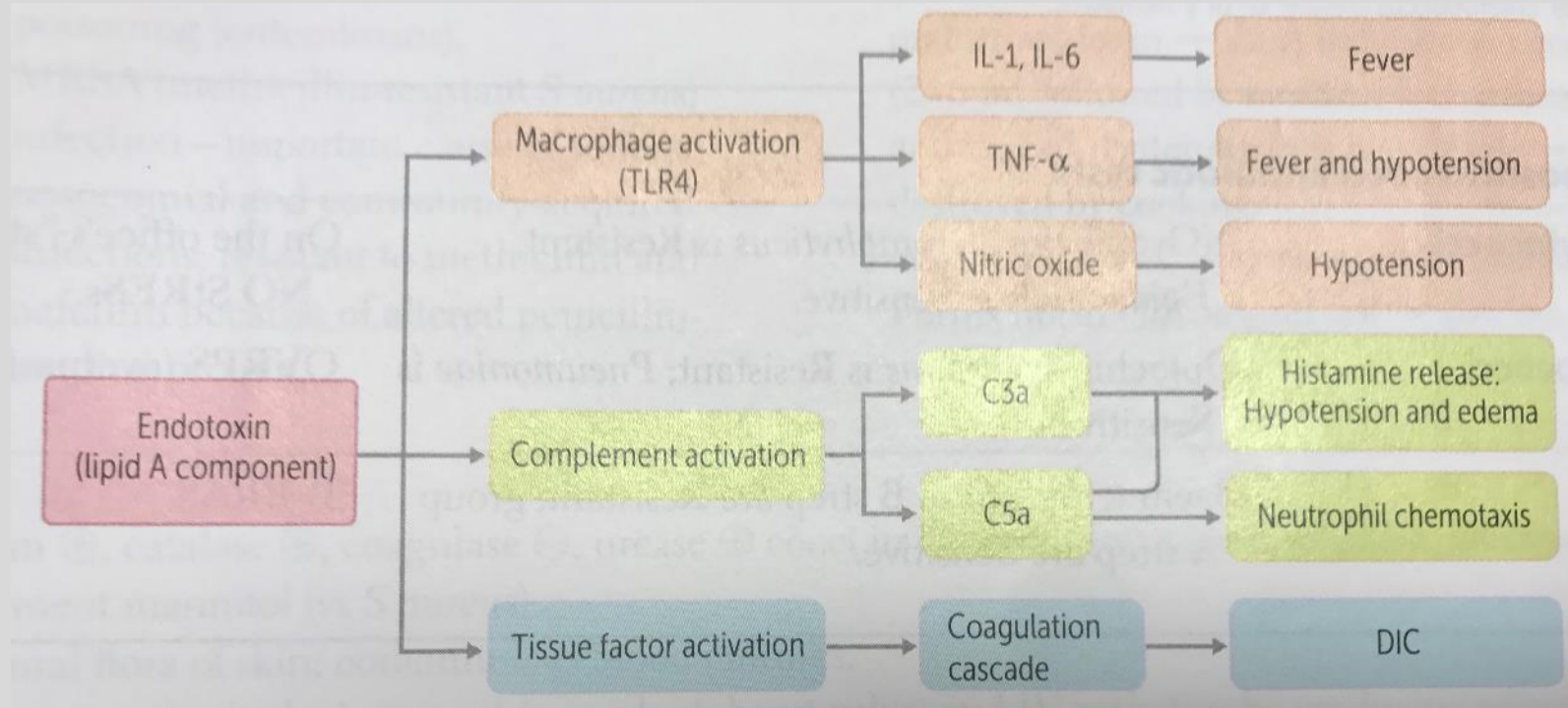
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Lipid A

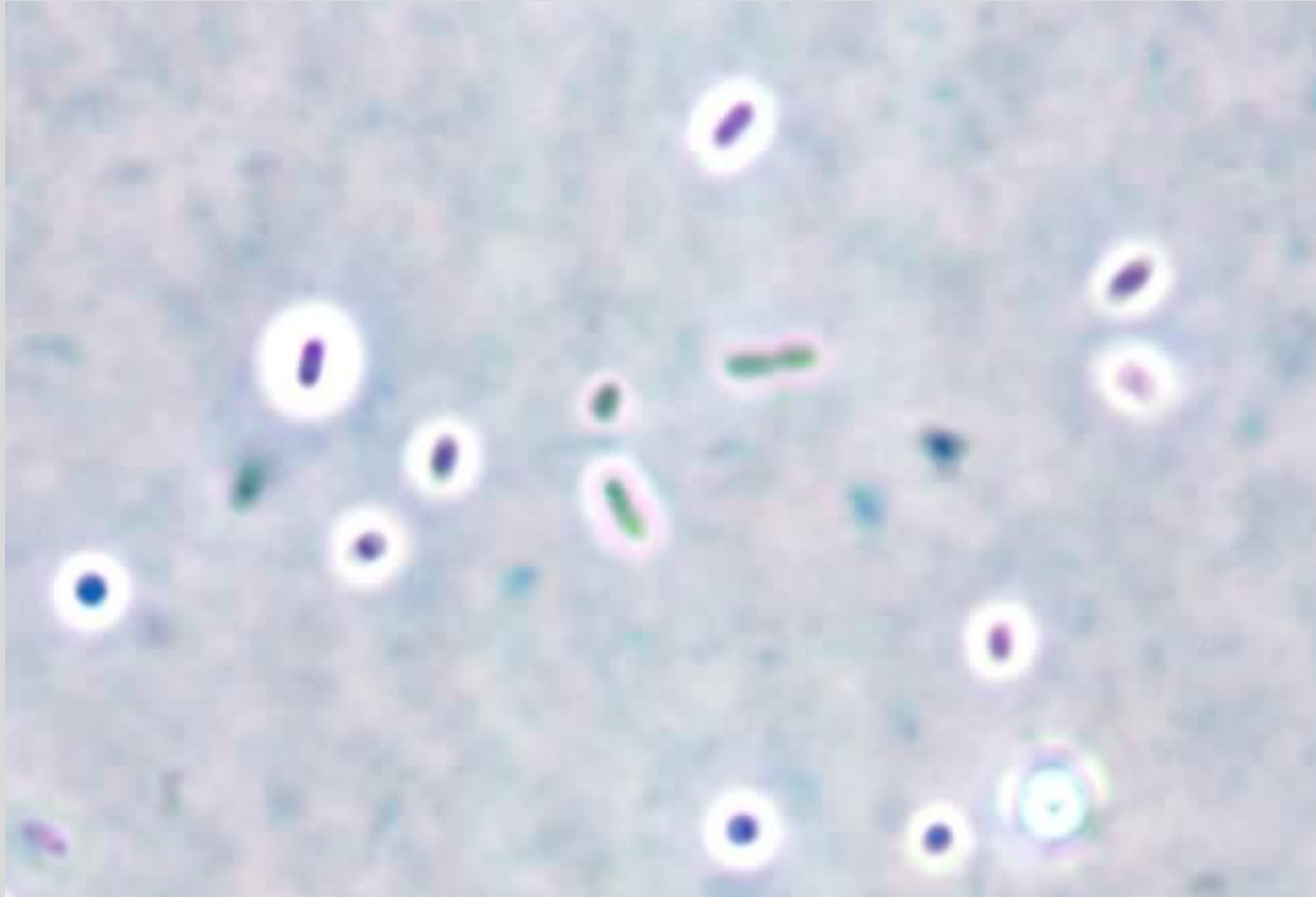


Tao, LE, V. Bhushan and M. Sochat. 2023. First Aid for the USMLE Step 1; McGraw Hill Publication.

Capsule

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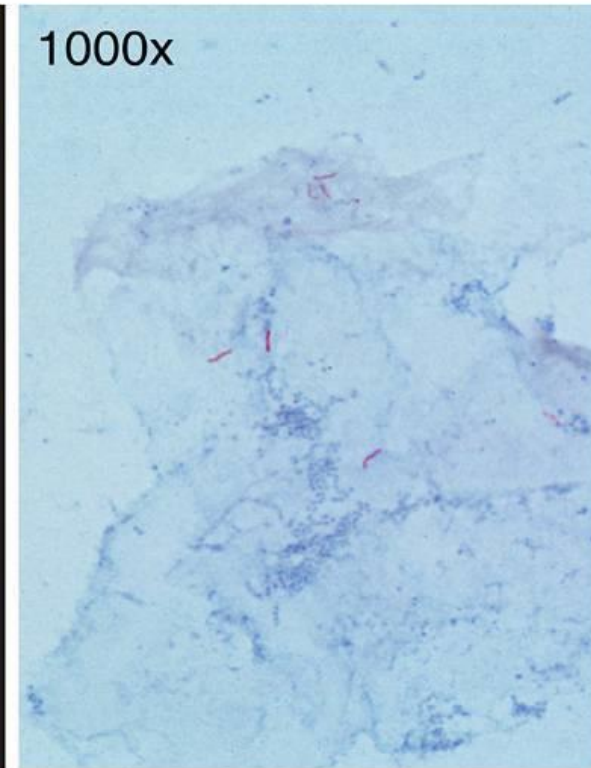
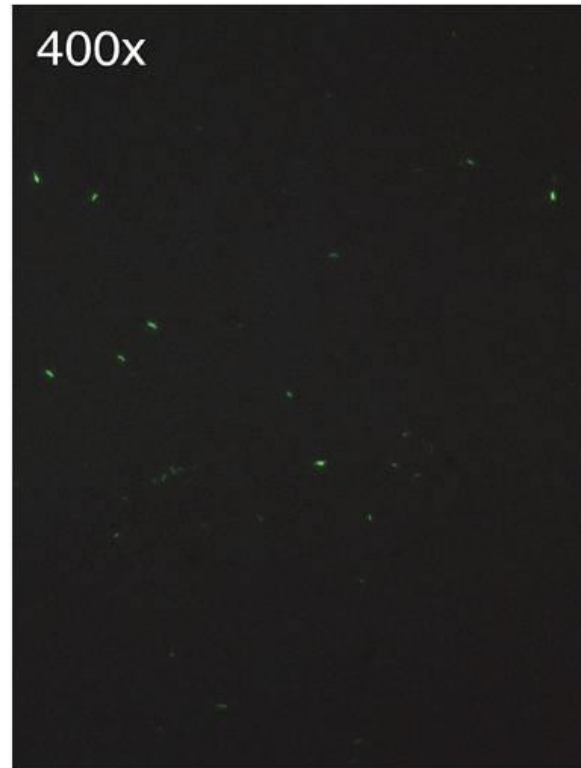
<http://microbeonline.com/wp-content/uploads/2016/10/Capsule-Stain.jpg>

Gram Nomenclature Exceptions/Challenges

- *Mycobacteria*
- *Mycoplasma*
- *Treponema*
- *Legionella, Rickettsia, Chlamydia, Bartonella, Ehrlichia & Anaplasma*

Acid Fast bacilli

- Ziehl-Neelsen stain
- Auramine-Rhodamine Fluorochrome staining:
 1. less time
 2. less experience
 3. lower magnification visualization (x400)
 4. Needs a fluorescent microscope
- Cultures/nucleic acid amplification need to be done

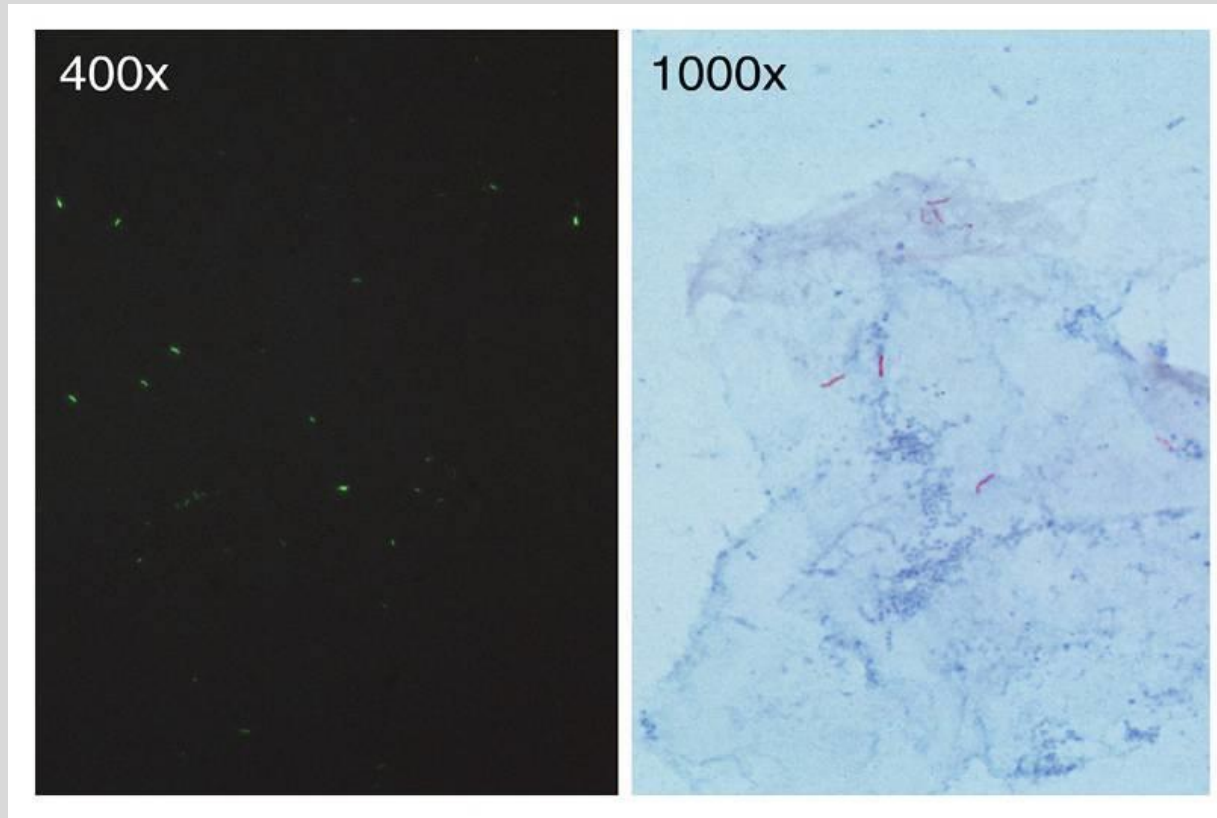


Source: Division of Laboratory Systems,
Public Health Practice Program Office, Center for Disease Control and Prevention (CDC)

Acid Fast bacilli

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Source: Division of Laboratory Systems,
Public Health Practice Program Office, Center for Disease Control and Prevention (CDC)

References

1. Murray, P.R., Rosenthal, K. S., and Pfaller, M.A. 2021. **Medical Microbiology**. Elsevier Inc.
2. Tao, LE, V. Bhushan and M. Sochat. 2023. **First Aid for the USMLE Step 1**; McGraw Hill Publication.
3. Engleberg, N. C., DiRita, V. and T.S. Dermody. 2013. **Schaechter's Mechanism of Microbial Disease**. Wolters Kluwer Health 5th ed. Lippincott Williams & Wilkins. ISBN: 978-0-7817-8744-4.

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>